

Let $f \in \mathbb{N} \rightarrow \mathbb{N}$ be the partial function $\{(x, 0) \mid \varphi_x(x) \uparrow\}$.

$$\text{Thus } f(x) = \begin{cases} 0 & \text{if } \varphi_x(x) \uparrow \\ \text{undefined} & \text{if } \varphi_x(x) \downarrow \end{cases}$$

f is not computable, because if it were, then $f = \varphi_e$ for some $e \in \mathbb{N}$ and hence

- if $\varphi_e(e) \uparrow$, then $f(e) = 0$ (by def. of f); so $\varphi_e(e) = 0$ (by def. of e), i.e. $\varphi_e(e) \downarrow$
- if $\varphi_e(e) \downarrow$, then $f(e) \uparrow$ (by def. of e); so $\varphi_e(e) \uparrow$ (by def. of f)

Contradiction! So f